SECOND BRAIN: AUTONOMOUS AI ENGINEERING PLATFORM

Comprehensive Engineering Fundamentals, Architecture Blueprint & AI Systems Guide

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1. Executive Summary & Problem Space

Modern AI researchers and high-output software engineers face severe *cognitive thrashing*: juggling multi-model experimentation benchmarks, complex academic manuscripts, circadian sleep debt, and fragmented knowledge repositories. Conventional productivity tools (Notion, Trello, Obsidian) fail because they are passive silos requiring high manual maintenance without automated verification.

Second Brain is architected as an *autonomous, event-driven AI Engineering Operating System & Distributed Multi-Agent Platform*. It standardizes agent tool interfaces via Anthropic's Model Context Protocol (MCP), eliminates semantic hallucinations using Corrective RAG (CRAG) + RAPTOR hierarchical tree summarization, extracts visual artifacts via Gemini 2.0 Flash VLM, calculates deterministic circadian burnout risks in <2ms via Classical ML (Isolation Forest), audits agent outputs with an automated LLM-as-a-Judge CI/CD Quality Flywheel (81.8/100 Composite Score), enforces a 5-layer OWASP LLM Defense Matrix, and anchors S2 thesis research on Instance-Dependent Cost-Sensitive Explainable AI.

2. Polyglot Architecture & Concurrency Model

A deliberate architectural choice is the **Polyglot Division of Concerns**, matching workload characteristics to language-level execution models:

Domain / Layer	Technology	Rationale & Architectural Benefit
AI Reasoning & Tool Fleet	Python 3.11 / FastAPI	Native integration with Google ADK, Anthropic MCP Server (SSE), PyTorch/CatBoost, scikit-learn, and async pgvector pipelines.
Background Telemetry Engine	Golang 1.22+ Worker (:8085)	True OS-thread concurrency via Goroutines & Channels. 10x lower memory footprint (~14MB) than Python workers; zero GIL lockups during heavy continuous cron evaluations.
Knowledge Vector Store	PostgreSQL + pgvector	Unified relational storage with 768-dimensional vector embeddings, HNSW graph indexing (m=16, ef_construction=64), and Row-Level Security (RLS).
Developer Cockpit Frontend	Vue 3 + Vite + Tailwind 3.0	Raycast-style Command Palette (Ctrl+K), Recruiter Quick Cockpit, animated SVG Radial Gauge, sub-second routing, and 100% PWA offline capability.

3. Standard Model Context Protocol (Anthropic MCP SSE)

To avoid proprietary tool lock-in, Second Brain exposes its core capabilities as an **Anthropic Model Context Protocol (MCP) Server** over Server-Sent Events (SSE) at /mcp/sse and /mcp/messages/. This enables external LLM desktop clients (such as Claude Desktop and Cursor IDE) to seamlessly consume Second Brain tools via standardized JSON-RPC 2.0:

- **8 Registered MCP Tool Schemas:** save_note, search_notes, get_thesis_progress, log_habit, record_focus_session, evaluate_burnout_radar, upload_visual_document, and run_rag_qa.
- **Google ADK Bidirectional Adapter:** Internal ADK agents can import MCP-compliant tools dynamically, facilitating modular multi-agent collaboration.

4. Next-Gen RAG: RAPTOR Knowledge Tree + Corrective RAG (CRAG)

Pure vector search suffers from semantic drift and technical token blindness. Second Brain implements a state-of-the-art **Hierarchical & Self-Corrective RAG Pipeline**:

- **RAPTOR Hierarchical Clustering:** Document embeddings are grouped into semantic clusters using **Gaussian Mixture Models (GMM)** and KMeans. Gemini recursively synthesizes high-level summary nodes, constructing a multi-tier Knowledge Tree for

macro-level conceptual retrieval.
• **Corrective RAG (CRAG) Routing:** Evaluates context confidence dynamically:
1. **CORRECT (> 0.70):** High precision; passes context directly to the generator.
2. **AMBIGUOUS (0.35 - 0.70):** Triggers **HyDE (Hypothetical Document Embeddings)** query expansion to retrieve missing contextual chunks.
3. **INCORRECT (< 0.35):** Activates Out-of-Distribution (OOD) guardrails, rejecting hallucinations with 100% adherence.
• **Cross-Encoder Reranker:** Re-scores Top-10 pgvector candidates before injecting into context windows, balancing dense similarity (0.50), lexical BM25 matching (0.30), and token overlap (0.20).

5. Multimodal Document Intelligence (Gemini 2.0 Flash VLM)

Technical engineering artifacts are rarely plain text. The POST /api/v1/notes/upload-visual endpoint leverages Gemini 2.0 Flash VLM to process complex visual documents:

• **Model Training Curve Analytics:** Analyzes training loss convergence, detects plateau/overfitting at specific epochs (e.g., CatBoost iteration 350), and extracts F1-Score / AUC-ROC metrics.
• **Handwriting OCR & LaTeX Transcription:** Converts handwritten lecture slides and mathematical formulas into standard monospace LaTeX expressions.
• **Automated Vector Ingestion:** Structured JSON outputs (summary, OCR, insights, tags) are vectorized into 768-D embeddings and linked to the RAPTOR tree in <50ms.

6. Classical ML Hybrid: Circadian & Burnout Anomaly Radar

A key indicator of engineering maturity is knowing when **not** to use Generative AI. Calculating real-time health anomalies on every telemetry heartbeat with an LLM is wasteful and slow (>500ms, costly tokens).

• **Isolation Forest in < 2ms:** Evaluates a 5-D feature vector (sleep_hours, sleep_debt, midnight_sessions, overdue_tasks_ratio, water_intake) in-memory with zero token spend.
• **Conditional Autonomous LLM Intervention:** When the Anomaly Score drops below the critical decision boundary (< -0.150), the system conditionally triggers Gemini 2.0 Flash to synthesize an empathetic, contextual energy recovery plan.
• **Animated SVG Radial Gauge:** Visualizes analog deflection with spring-physics needle rotation and real-time threshold alert.

7. Production Evals & CI/CD Observability (Quality Flywheel)

Every code commit is continuously audited against a 20-case curated Golden Dataset across 7 academic domains using **LLM-as-a-Judge RAG Triad** scoring with sub-1.2s live spot-check capability:

Evaluation Metric	Benchmark Score	CI/CD Gate Threshold	Status
Faithfulness (Groundedness)	**85.0% (0.850)**	>= 80.0%	**PASSED**
Context Recall	**86.0% (0.860)**	>= 80.0%	**PASSED**
Answer Relevancy	**78.5% (0.785)**	>= 75.0%	**PASSED**
Out-of-Distribution Refusal	**100.0% (1.000)**	>= 90.0%	**PASSED**
Composite AI Quality Index	**81.8 / 100**	>= 75.0 / 100	**EXCELLENT**

8. S2 Master's Thesis: Instance-Dependent Cost-Sensitive Explainable AI (FMIPA UGM)

Title: *Stabilisasi Penjelasan Model Instance-Dependent Cost-Sensitive Menggunakan Regularisasi dan Uji Permutasi Biaya Terstratifikasi*
Candidate: Frans Alwan Purba (NIM: 25/563545/PPA/07116) · **Supervisors:** Afiahayati, S.Kom., M.Cs., Ph.D & Prof. Dra. Sri Hartati, M.Sc., Ph.D.
Problem & Novelty: Ballegeer et al. (2025, EJOR) showed that optimizing models for Average Expected Cost (AEC) destabilizes local SHAP/LIME explanations (CoV > 0.25). Our work addresses this via a rigorous 3-stage methodology:
• **Stage 1:** Negative control noise floor estimation across 32 HMEQ configurations (N=5,960, L=25 bootstrap).
• **Stage 2 (Dual-Arm):** Arm 1 applies model weight regularization with a non-degeneracy filter ($\sigma > 0.01$); Arm 2 performs stratified counterfactual cost-permutation (cost-shuffle across 5 seeds).
• **Stage 3:** Sign-Flip permutation testing (B=2,000 + Benjamini-Hochberg FDR) and No-Lose Decision Matrix covering Scenarios A, B, and C with fast 140ms permutation simulation.

9. 5-Layer AI Governance & OWASP LLM Defense Matrix

The system incorporates an end-to-end defense-in-depth security model compliant with the **OWASP Top 10 for Large Language Models**:

• **Layer 1 (LLM01 Prompt Injection):** Enforces strict XML delimiters (<user_query>) and pre-flight heuristic pattern scanning to block instruction overrides.

• **Layer 2 (LLM04 Model DoS):** Token-bucket rate limiting via FastAPI (max 20 RAG/min, max 5 VLM/min) preventing Denial of Wallet.

• **Layer 3 (LLM02 Sensitive Data Exposure):** Presidio-style regex PII auto-masking (< 2ms) redacting phone numbers (+62/08xx), student NIM, emails, and API keys.

• **Layer 4 (LLM06 Excessive Agency):** Strict separation of read-only vs protected write tools in MCP tool routing.

• **Layer 5 (Auditability):** Append-Only PostgreSQL Row-Level Security audit logs preventing historical tampering.

10. UI/UX 3.0 Engineering & Architectural Decision Records (ADRs)

Designed to Raycast/Linear developer ergonomics standards: **Global Command Palette (Ctrl+K)** for instant power-user actions, **Sticky Recruiter Quick Cockpit** for 1-click scenario showcase, **AI Execution Trace Waterfall** with sub-millisecond latency accounting, and formal documentation: 4 Architectural Decision Records (ADR-001 to ADR-004), 2 Model Cards, and an automated Eval Reproducibility Guide.